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Foundations of Programming

Assignment No. : 10

PROBLEM STATEMENT:

Write a program in C to simulate a simple calculator that performs basic arithmetic operations such as addition, subtraction, multiplication, and division. The calculator should also perform special operations like computing x^y (power) and x! (factorial).

OBJECTIVE:

- To understand the use of switch-case statements in C.
- To perform basic arithmetic and special mathematical operations.
- To enhance problem-solving skills using menu-driven programs.

THEORY:

A calculator is a basic computing device used to perform arithmetic operations. In C programming, a simple calculator can be implemented using conditional or switch-case statements. Arithmetic operations like addition, subtraction, multiplication, and division are fundamental operations. Special operations such as power (x^y) and factorial (x!) require loops for repeated calculations.

PLATFORM:

Linux

ALGORITHM:

C PROGRAM:

```
#include <stdio.h>

int main()

{

   int choice;

   int a,b;

   int i;

   long long fac1,fac2;

   double power=1;
```

```
printf("Arithmetic Menu\n");
printf("1. Addition\n");
printf("2. Subtraction\n");
printf("3. Multiplication\n");
printf("4. Division\n");
printf("5. Power (x^y)\n");
printf("6. Factorial (x! and y!)\n");
printf("Enter your choice: ");
scanf("%d",&choice);
printf("Enter two integers (x y): ");
scanf("%d %d",&a,&b);
switch(choice)
{
   case 1:
      printf("Result = %d\n",a+b);
      break;
   case 2:
      printf("Result = %d\n",a-b);
      break;
   case 3:
      printf("Result = %d\n",a*b);
      break;
   case 4:
      if(b!=0)
         printf("Result = %.2f\n",(double)a/b);
      else
         printf("Division by zero not allowed.\n");
```

```
    break;
  case 5:
    power=1;
    if(b<0)
      printf("Negative powers not supported.\n");
    else
    {
      for(i=0;i<b;i++)
        power=power*a;
      printf("Result = %.2f\n",power);
    }
    break;
  case 6:
    if(a<0 || b<0)
    {
      printf("Factorial not defined for negative numbers.\n");
      break;
    }
    if(a>20 || b>20)
    {
      printf("Numbers too large.\n");
      break;
    }
    fac1=1;
    fac2=1;
    for(i=1;i<=a;i++)
      fac1=fac1*i;
```

```
        for(i=1;i<=b;i++)
            fac2=fac2*i;
        printf("Factorial of x = %lld\n",fac1);
        printf("Factorial of y = %lld\n",fac2);
        break;
    default:
        printf("Invalid choice.\n");
  }
  return 0;
}
```

```c
#include <stdio.h>
int main()
{
    int choice;
    int a,b;
    int i;
    long long fac1,fac2;
    double power=1;
    printf("Arithmetic Menu\n");
    printf("1. Addition\n");
    printf("2. Subtraction\n");
    printf("3. Multiplication\n");
    printf("4. Division\n");
    printf("5. Power (x^y)\n");
    printf("6. Factorial (x! and y!)\n");
    printf("Enter your choice: ");
    scanf("%d",&choice);
    printf("Enter two integers (x y): ");
    scanf("%d %d",&a,&b);
    switch(choice)
    {
        case 1:
            printf("Result = %d\n",a+b);
            break;
        case 2:
            printf("Result = %d\n",a-b);
            break;
        case 3:
            printf("Result = %d\n",a*b);
            break;
        case 4:
            if(b!=0)
                printf("Result = %.2f\n",(double)a/b);
            else
                printf("Division by zero not allowed.\n");
            break;
        case 5:
```

INPUT:

User selects an operation from the menu and enters required values.
Example Input:
Choice: 1
Numbers: 10 20

```
Arithmetic Menu
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Power (x^y)
6. Factorial (x! and y!)
```

Screen Shot of Input

OUTPUT:

```
Enter your choice: 3
Enter two integers (x y): 2 90
Result = 180
```

Example Output:
Result = 30.00

Screen Shot of Output

CONCLUSION:

Thus, a simple calculator program was successfully implemented using C programming. The program demonstrates the use of switch-case statements, loops, and arithmetic operations.

FAQs:

1. What is a simple calculator program?
2. Why is switch–case preferred in calculator programs?
3. How does the program handle division by zero?
4. How is the power operation (x^y) implemented without using math library functions?
5. What is factorial and how is it calculated using loops in C?